ISLAMIC UNIVERSITY OF TECHNOLOGY

Lab Report 3

Livestock Management System

CSE 4408: System Analysis and Design Lab

Lab 3: Project Management

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Table of Contents

1. Problem Definition and Project Selection

1.1 Organizational Context

The project concerns an Integrated Smart Farm System covering poultry, fisheries, and cattle. The target organization currently depends on manual records and disconnected operational routines, while the proposed Livestock Management System will provide a centralized web-based application supported by a relational database. The overall goal is to improve efficiency, sustainability, resource sharing, tracking, and decision-making across all livestock subsystems.

1.2 Core Problem Statement

The existing farm operation is slowed down by paper-based record keeping, isolated subsystem data, delayed reporting, and weak tracking of feed, inventory, health, and production activities. Because data is recorded manually, the farm faces data loss, human error, duplicate work, inconsistent records, missed alerts, irregular vaccination follow-up, overstocking or shortage of resources, and weak accountability.

Issue	Objective	System Requirement
Manual records and data loss	Reduce dependency on paper logs	Auto data entry support and report generation
Disconnected poultry, fisheries, and cattle subsystems	Create one shared information base	Centralized relational database supporting all subsystems
No real-time monitoring	Provide timely operational visibility	Monitoring and alert system with dashboard analytics
Delayed decisions	Support faster management decisions	Real-time tracking and summarized reports
Poor resource tracking	Improve feed, inventory, and health tracking	Unified interface with synchronized subsystem data

Table 1: Issues, Objectives, and Requirements

1.3 System Scope and Constraints

The proposed system will be software-focused. It will cover livestock inventory, daily logs, feed and resource tracking, health monitoring, reporting, and basic inventory updates. It will not include physical IoT automation, GPS logistics, e-commerce marketplace functions, payroll, or external regulatory reporting in the first release. This keeps the project achievable for a student systems analysis and design context.

- Budget constraint: the first-year estimated cost is limited to BDT 510,000.
- Technology constraint: the current organization has no existing software infrastructure, so the system must be built from scratch.
- Adoption constraint: rural farm workers may need training and a simple interface to shift from paper-based routines.
- Connectivity constraint: the design should consider possible network or electricity interruptions.

1.4 Project Selection Against the Five Textbook Criteria

Criterion	Assessment for Livestock Management System
Management backing	The project has strategic value for the owner because it centralizes farm data and improves reporting. Stakeholder input was available through a conventional livestock farm contact.
Appropriate timing	The farm is still manual and paper-based, so the change is timely before the operational scale becomes harder to control.
Strategic goal improvement	The system directly supports efficiency, sustainability, better tracking, and improved decision-making.
Practical resources	Required technologies exist: web/mobile interface, dashboards, reporting tools, and relational databases. Missing expertise can be hired.
Worthwhile compared to alternatives	Keeping the current system preserves data loss and delay. A packaged solution would need heavy customization. A custom web-based system is the most aligned option.

Table 2: Project Selection Justification

2. Project Overview and Data Sources

The selected solution is to replace the manual system with a web-based livestock management application using relational databases. The system should connect resource records, animal health data, daily work schedules, inventory updates, and reporting in one synchronized environment.

Data Area	Source / Basis
Farm operation context	Direct line to a conventional livestock farm stakeholder
Software development cost	Approximate cost input from the Head of IT of Genex Infosys Ltd.
Farm resource and operation cost	Estimated from current manual record-keeping process
Development and training cost	Estimated for system analysis, development, database setup by the Head of IT of Genex Infosys Ltd.
Livestock feed prices	Collected from ARMAN Feeds and Fisheries Ltd. as stated in the presentation

Table 3: Data Sources and Estimation Basis

3. Preliminary Feasibility Assessment

3.1 Technical Feasibility

The current farm has no existing software infrastructure, so the project cannot be treated as a small enhancement. A fresh system is required. However, the required technologies are common and available: a web/mobile interface, dashboards, reporting tools, and relational databases. The lack of in-house expertise is a manageable limitation because UI development, backend development, testing, and mobile development can be hired or outsourced.

Packaged software may exist, but it would require customization because poultry, fisheries, and cattle data must be synchronized in a farm-specific way. Therefore, the project is technically feasible if it is built as a modular custom system with clear subsystem integration points.

3.2 Economic Feasibility

The economic case is based on a one-time first-year investment and recurring yearly benefits. Major cost items include system analysis, design, outsourced development, database/server setup, employee training, and transition support. The main benefits are reduced manual record-keeping, lower feed mismanagement, faster reports, better inventory control, accountability, and more reliable transaction records.

The capital expenditure of the system is estimated to be around BDT 16,72,716 (see the end of the report for an attached xlsx document for a detailed breakdown of the estimation). Furthermore, an annual maintenance contract (AMC) is included for the system which is taken to be 20% of the capital expenditure of system. The AMC is, thus, calculated to be BDT 394,543, which is to be paid annually starting from the second year.

The yearly revenue of the system is estimated to be around BDT 492,000, taking accounts for all the subsystems, ie, the cattle farm, the poultry farm and the fishery.

The first-year result shows a loss of BDT12,40,716, but from the second year onward the system generates a strong positive return of BDT 97,457 because the initial setup cost is not repeated at the same level. This makes the system economically feasible over more than one year of operation. The system is exepected to break even at 17.16 years or 206 months.

3.3 Operational Feasibility

Operational feasibility depends on whether the system will work inside the farm environment and whether users will actually use it. The proposed system improves workflows by replacing scattered paper logs with digital records, centralized reports, daily logs, worker schedules, inventory updates, and health monitoring. User feedback and training are important because farm workers may not be used to digital tools.

The system is operationally feasible if the interface is kept simple, the training is hands-on, and the deployment is gradual. A local-language interface, mobile-friendly screens, and clear confirmation messages should be used to reduce resistance.

Feasibility Area	Positive Factors	Main Risk	Conclusion
Technical	Available web/mobile, dashboard, reporting, and database technology	No existing software; custom build needed	Feasible
Economic	Yearly benefit is close to the initial cost and continues after first year	First year has a small net loss	Feasible
Operational	Simple digital workflow improves logs, schedules, inventory, and health records	Possible resistance from manual-system users	Feasible (with training)

Table 4: T.E.O. Feasibility Summary

4. Workload Estimates and Before-After Projection

Workload estimation supports both technical and economic feasibility. The proposed system reduces manual effort, duplicate entry, and reporting delay, while increasing the volume of structured records that can be searched and summarized automatically.

Work Area	Current Manual Workload	Proposed System Workload	Expected Effect
Daily feed logging	Workers manually write feed usage in registers; data may be incomplete or duplicated.	Workers submit batch-wise feed logs through forms.	Less duplicate work and better inventory depletion tracking.
Inventory tracking	Stock levels are checked manually and updated late.	Inventory updates are stored centrally after purchases and usage.	Earlier shortage detection and better control.
Health monitoring	Vaccination and treatment notes may stay in separate paper records.	Veterinary and health records are linked to livestock batches.	Fewer missed alerts and better treatment history.
Reporting	Owner waits for manually prepared summaries.	Reports and dashboards are generated from stored records.	Faster decision-making and less managerial time.
Subsystem coordination	Poultry, fishery, and cattle operations exchange information manually.	Subsystems use synchronized data and shared resource records.	Improved resource planning and accountability.

Table 5: Before-and-After Workload Projection

Record Type	Estimated Frequency	Main User
Feed logs	Daily for poultry, fisheries, and cattle batches	Farm Worker
Inventory updates	Whenever stock is used or replenished	Manager / Worker
Health records	At vaccination, examination, or treatment events	Veterinarian / Manager
Task schedules	Daily or weekly	Manager
Reports	Weekly, monthly, or on demand	Owner / Manager

Table 6: Estimated Transaction Volume

The technical workload is acceptable because the main records are text, numbers, dates, and status values. Even with daily records across all three subsystems, a relational database can handle the expected storage and query load. The economic workload also improves because the system reduces repeated manual recording and summary preparation.

5. Cost-Benefit Forecast

5.1 Cost Forecast

The cost is estimated in the attached xlsx document, taking the Manday rate to be BDT 7,500. The “CapEx” worksheet provides the details on the factors considered and their estimated Mandays required, The “Development Expense” worksheet provides a detailed breakdown of the numerous factors related to the actual coding part of the development lifecycle. The summary of the cost analysis is show here.

Cost Item	Estimated Mandays	Estimated Cost (BDT)
Requirement Analysis	10.88	81,600
Feasibility Study	4.17	31,275
System Design	27.00	202,500
Development / Coding	131.87	989,025
Testing	19.83	55,000
Training and Transition	5.17	38,775
Deployment / Implementation	13.50	101,250
Total first-year cost	212.41	1,593,069
Total first-year cost + 5% contengency		**1,672,716**

Table 7: Capital Expenditure

5.2 Benefit Forecast

Benefit Item	Basis	Estimated Benefit (BDT/year)
Reduced manual record-keeping effort	(BDT 30,000 manager effort - BDT 15,000 data insertion) x 12 months	180,000
Reduced cattle feed mismanagement	10 x 25 kg cattle feed packets x 12 months for around 20 cows	140,000
Reduced poultry feed mismanagement	2 x 50 kg chicken feed sacks x 12 months for around 1,000 chickens	60,000
Reduced fish feed mismanagement	5 x 25 kg fish feed sacks x 12 months for around 5-acre mixed carp pond	100,000
Faster reporting and decision-making	Estimated 5% operational upscaling effect	12,000
Better inventory control	Reduced overstocking/shortage and improved stock visibility	10,000
Total yearly benefit		**492,000**

Table 8: Yearly Benefit Forecast

5.3 Tangible and Intangible Benefits

Category	Items
Tangible benefits	Reduced labor effort, reduced feed waste, better inventory control, faster reporting value
Intangible benefits	Improved data accuracy, stronger accountability, better decision confidence, lower risk of missed vaccination or health follow-up
Tangible costs	Analysis, development, database/server setup, training, transition
Intangible costs/risks	Temporary learning curve, possible resistance, transition disruption, dependency on power/network availability

Table 9: Tangible and Intangible Cost-Benefit View

6. Break-Even and Payback Analysis

The cost-benefit comparison uses BDT 510,000 as the initial investment and BDT 492,000 as the expected yearly benefit. The monthly benefit is calculated as BDT 492,000 / 12 = BDT 41,000.

Item	Amount
Initial / first-year project cost	BDT 510,000
Expected yearly benefit	BDT 492,000
Monthly benefit	BDT 41,000
First-year net position	BDT 18,000 loss
Second-year cumulative position	BDT 474,000 gain

Table 10: Cost-Benefit Summary

6.1 Payback Period

Payback period measures the time needed for accumulated benefits to recover the initial investment.

Payback Period = Initial Investment / Annual Benefit = 510,000 / 492,000 = 1.0366 years.

In months, 1.0366 x 12 = 12.44 months. Therefore, the system is expected to recover its first-year investment in about 12.5 months, or approximately 13 months.

6.2 Break-Even Point

Break-even occurs when total benefits become equal to total cost.

Break-Even Point = Initial Investment / Monthly Benefit = 510,000 / 41,000 = 12.43 months.

The break-even point is therefore approximately 13 months. After this point, the accumulated financial benefits exceed the estimated initial cost.

7. Financial Estimation Spreadsheets

A detailed look into the financial estimations that were done in the making of this report is presented here.